The ECG Towel

A Concept for Pre-Hospital High-Density Cardiac Mapping

Aaron Garcia

aaron@garcia.ltd

Version 7.0 | PUBLIC

8 August 2026

1.0 Two Minutes at the Roadside

A driver has pulled onto the verge of an A-road, grey and sweating, with palpitations. The paramedic who reaches him has one diagnostic window and one tool: a 12-lead ECG, twelve electrical viewpoints on a heart that is misbehaving in three dimensions. If the arrhythmia is transient it may not even appear in the strip. If it is atypical, the pattern may not be recognisable in a lay-by at night. The patient will be driven to a hospital where, hours later, an electrophysiology lab could map his heart with hundreds of virtual electrodes and locate the misfiring circuit precisely. The information gap between the verge and that lab is the subject of this paper.

The concept is a rapid-deployment sensing wrap—informally, an ECG towel: a flexible, high-density electrode array a paramedic could throw around a patient's chest in a couple of minutes [TARGET], which maps the heart's electrical activity in 3D and sends the result ahead to the receiving hospital. No CT scan, no precise electrode placement, no specialist on scene.

None of this hardware exists. This is a concept paper in the strict sense: it describes an idea, tests it against published work, and specifies what would have to be true for it to survive. The previous version of this document was an investment memorandum; it has been withdrawn, for reasons given in Section 5.

2.0 Why This Is Nearly Possible

Three developments make the idea worth examining now rather than dismissing as science fiction.

2.1 Imageless Cardiac Mapping Is Real

Electrocardiographic imaging (ECGI) reconstructs the heart's surface electrical activity from many body-surface electrodes. For years its practical weakness was the requirement for a CT or MRI scan to establish torso and heart geometry—Medtronic's CardioInsight system, the commercial reference point with 252 electrodes, works this way. That constraint has now fallen: Corify Care's ACORYS system performs imageless ECGI, fitting a statistical shape model to a surface torso scan instead of tomography, with 63 electrodes, published clinical validation, and US regulatory clearance (Europace 2026; PubMed 40293957). The hardest scientific prerequisite for pre-hospital cardiac mapping—geometry without a scanner—has been demonstrated by others, in the calm of a clinic. This paper builds on that foundation; it does not claim it.

2.2 Pre-Hospital ECG Already Saves Lives

Recording a 12-lead ECG before hospital arrival is standard practice in developed ambulance services, and it measurably improves outcomes: patients with a pre-hospital ECG reach primary coronary intervention faster and show lower mortality (see PMC8312365, 2021, and the review literature it cites). The clinical pathway for "diagnose earlier, treat sooner" is proven; what is under question is only how much more diagnosis the roadside can support. Notably, even 12-lead placement accuracy by trained paramedics is imperfect, which is an argument for, not against, systems that tolerate imprecise placement.

2.3 Flexible High-Density Arrays Are Emerging

Conformable electrode arrays dense enough for ECGI are appearing in the materials literature—conducting-polymer eutectogel arrays on flexible substrates are an example (Serrano et al., 2024)—though they remain laboratory demonstrations. The same literature is honest about the field's core problem: ECGI reconstruction accuracy is still contested, and that debate transfers in full to anything proposed here. Higher electrode density may improve reconstruction; that it does so decisively is not yet established.

2.4 What Nobody Has Done

Searches behind this paper found no high-density ECGI system designed for, or trialled in, pre-hospital use: no ambulance deployment, no rapid-application form factor, no motion-tolerant reconstruction pipeline. Imageless ECGI exists in the clinic. High-density flexible arrays exist in the lab. Pre-hospital 12-lead exists everywhere. The combination—clinic-grade mapping arriving with the ambulance—appears to be unclaimed territory, and that combination is this paper's entire subject.

3.0 The Concept

The proposal, stated compactly: a flexible chest wrap carrying several hundred electrodes [TARGET], applied over or around existing clothing contact points in under two minutes [TARGET], tolerant of partial contact because density provides redundancy [HYPOTHESIS], establishing its own geometry through a statistical shape model constrained by body measurements and inter-electrode impedance [HYPOTHESIS, following the imageless approach validated by ACORYS], with signal processing split between an on-vehicle unit and the cloud, and results returned as confidence-gated decision support—never as an autonomous diagnosis.

Three design choices deserve explanation. The towel form factor, rather than a fitted vest, is chosen for speed and size-independence: a wrap with excess sensors accommodates any chest without sizing, and the system discards non-contacting channels rather than requiring all of them. The electrode count—the original concept specified 700—should be read as "as many as manufacturing allows" rather than a derived optimum; whether hundreds of electrodes outperform ACORYS's 63 in reconstruction accuracy is an open research question the concept inherits from the literature, not a settled advantage. And the output is deliberately constrained: a confidence-scored map and rhythm classification for the receiving hospital, with anything below a confidence threshold collapsing to "use standard care"—a fail-safe posture that earlier versions of this work developed in detail and which survives as the concept's regulatory spine.

What the paramedic sees is a traffic-light state, not electrophysiology: apply the wrap, confirm green, continue standard care while the system works. The cognitive load target is “no new interpretation burden”; whether that is achievable is a human-factors question for any future prototype.

4.0 The Hard Problems, In Order

An honest concept paper is mostly a list of ways the concept could die. In descending order of lethality:

4.1 Motion Noise

Everything published on high-density ECGI comes from stationary patients in controlled rooms. An ambulance is a vibrating, accelerating Faraday-hostile environment wrapped around a moving patient with active chest muscles. Whether micro-volt cardiac signals can be recovered across hundreds of channels under those conditions is completely unevidenced—no study found in the searches behind this paper addresses it. This is the gating question. If roadside and in-transit noise cannot be beaten, the concept is dead regardless of every other merit, and the first experiment anyone should run (Section 6) attacks exactly this.

4.2 Does Density Pay?

ACORYS achieves clinical utility with 63 electrodes. The concept assumes that many more electrodes plus partial-contact tolerance beats fewer, well-placed ones. The materials literature suggests density could improve reconstruction; it has not demonstrated that it does, and there is a plausible opposing view that reconstruction accuracy is limited by the inverse problem’s conditioning, not by electrode count. If 63 careful electrodes equal 500 careless ones, the towel loses its reason to exist and the field should simply ruggedise existing vests.

4.3 Contact, Skin, and Bodies

Hair, sweat, body habitus, and movement will defeat some fraction of electrodes on every application, and the failures will cluster spatially rather than scatter randomly. The concept survives only if diagnostic-critical regions retain coverage after realistic contact loss—a testable claim, but one whose test requires hardware that does not exist. Demographic validity belongs in the same category: any geometry model and any classifier must be validated across body types, sexes, and ethnicities before clinical claims are made, and a system unvalidated for a subgroup must say so on its face rather than guess.

4.4 The Decision-Support Boundary

A device that tells a paramedic anything at all becomes part of clinical decision-making, with the liability and regulatory weight that follows (EU MDR Class IIb territory; UK and US analogues). The concept’s posture—confidence-gated output, defaulting to standard care, never overriding clinical judgement—is designed to keep the device advisory. Whether regulators accept that framing for a novel pre-hospital mapping device is untested, and the regulatory pathway would be long. This paper takes no position on commercialisation; the concept is published openly precisely so that anyone, including incumbent manufacturers, can pursue it.

5.0 What This Paper Is Not

There is no device, no prototype, no dataset, no company, no funding request, and no clinical claim here. Nothing in this document has been built or measured by its author.

The version this paper replaces was styled as an investment memorandum seeking seed funding. It has been withdrawn because, alongside its legitimate engineering analysis, it contained personnel, advisory, and market-diligence claims that had no basis in reality and should never have survived review. No advisors were engaged, no letters of intent obtained, no applications submitted, no interviews conducted. The author has contacted no one regarding this work. This replacement carries only what can be defended: an idea, its published foundations, and its open problems.

6.0 How to Prove It or Kill It

The concept reduces to three experiments, each cheap relative to what it settles, each with a kill condition. They are stated as proposals—nothing here has been started.

#	Experiment	Question settled	Kill condition
1	Bench array in vibration: a small multi-channel flexible array recording a cardiac phantom (then volunteers) on a shake table and in a moving vehicle	Can micro-volt signals survive ambulance-grade motion across many channels?	If adaptive filtering cannot recover diagnostic-band signal at realistic vibration, the concept dies here
2	Density study in software: reconstruction accuracy versus electrode count and contact fraction, using published BSPM recordings where access permits, plus simulated dropout	Do hundreds of imperfect contacts beat tens of good ones?	If accuracy plateaus at low electrode counts, the towel form factor loses its rationale
3	Imageless geometry under field constraints: shape-model heart localisation using only measurements obtainable at a roadside	Does the ACORYS-class approach survive without a controlled torso scan?	If geometry error exceeds what reconstruction tolerates, pre-hospital imageless mapping is premature

A research group with ECGI experience could run experiments 2 and 3 with no new hardware at all. Experiment 1 needs a bench rig. If all three survive, the case for building a prototype writes itself; if any fails, this paper has still done its job by marking the grave accurately.

7.0 Invitation

This concept is published openly, MIT-licensed, for anyone to attack, adopt, or extend—ECGI researchers who can say whether density pays, signal-processing engineers who know what vibration does to micro-volt channels, paramedics who know what survives contact with a real roadside, and regulators who can say where decision support ends. Criticism is more useful than endorsement. If the idea is wrong, the fastest possible demonstration of that is a contribution.

Contact: aaron@garcia.ltd

Appendix A: Document Metadata

Document Title	The ECG Towel
Subtitle	A Concept for Pre-Hospital High-Density Cardiac Mapping
GUID	AG-2026-0808-2941
Version	7.0
Date	8 August 2026
Author	Aaron Garcia
Affiliation	Independent Researcher
Contact	aaron@garcia.ltd
Origin Context	Rewritten from the author's original 2024 concept; replaces a withdrawn investment memorandum
Classification	PUBLIC
Status	RELEASED

Appendix B: Glossary

Term	Definition
BSPM	Body Surface Potential Mapping — recording cardiac signals from many torso electrodes
CT	Computed Tomography — X-ray-based 3D imaging
ECG	Electrocardiogram — recording of the heart's electrical activity
ECGI	Electrocardiographic Imaging — reconstruction of heart-surface electrical activity from body-surface recordings
EP	Electrophysiology — the study and treatment of the heart's electrical system
Inverse problem	Inferring heart-surface signals from body-surface measurements; mathematically ill-conditioned
MDR	Medical Device Regulation (EU) 2017/745
SSM	Statistical Shape Model — a mathematical model of anatomical variation across a population
STEMI	ST-Elevation Myocardial Infarction — a heart attack requiring urgent intervention

Appendix C: Methodology and Transparency Statement

Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The analytical framework reflects professional history, focusing on practical application.

Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools — specifically Claude, Google Gemini, and ChatGPT — to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial accessibility. As a writer with dyslexia, I leverage these models as assistive technology to refine grammar and streamline sentence structure.

Verification and Integrity

While AI assisted in processing data, the intellectual oversight is entirely human. I performed manual validation of all citations and accept full responsibility for the accuracy and originality of the final output.

Appendix D: References

Citation	Reference
Europace 2026	“Validation of an imageless electrocardiographic imaging technique for the non-invasive mapping of regular atrial tachyarrhythmias.” EP Europace, 28(4), 2026.
PubMed 40293957, 2025	“Real-Time Cardiac Mapping with a Noninvasive Imageless Electrocardiographic Imaging System.” PubMed 40293957, 2025.
CinC 2022	“Validation of a Novel Imageless Non-Invasive Electrocardiographic Imaging.” Computing in Cardiology, 2022.
Corify FDA	“Corify Care picks up FDA nod for cardiac mapping tech.” MassDevice, 2026.
Serrano et al., 2024	“High Density Body Surface Potential Mapping with Conducting Polymer-Eutectogel Electrode Arrays for ECG imaging.” Advanced Science, 2024.
PMC8312365, 2021	“Accuracy of ECG chest electrode placements by paramedics: an observational study.” PMC8312365, 2021.
ED BSPM 2007	“Estimated body surface potential maps in emergency department patients with unrecognized transient myocardial ischemia.” PubMed 17993313, 2007.
CardioInsight	Medtronic CardioInsight Noninvasive 3D Mapping System (252-electrode, CT-based). Product reference.